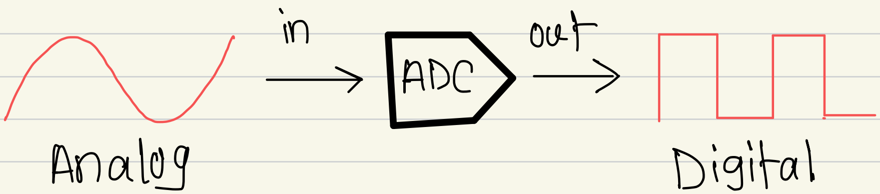
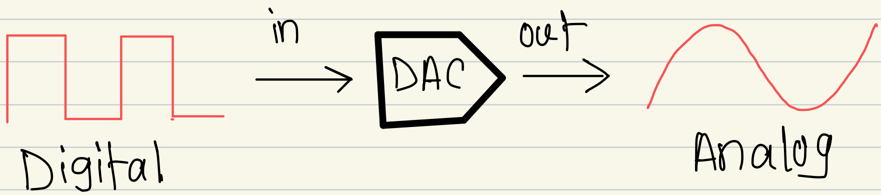
(ADC & DAC)

ADC → Analog to Digital Conversion

DAC → Digital to Analog Conversion





Analog means data could take any real value at each instant of time. It is continuous in time and amplitude. All physical world's signals informations are analog (Temp, pressure, sound ---)

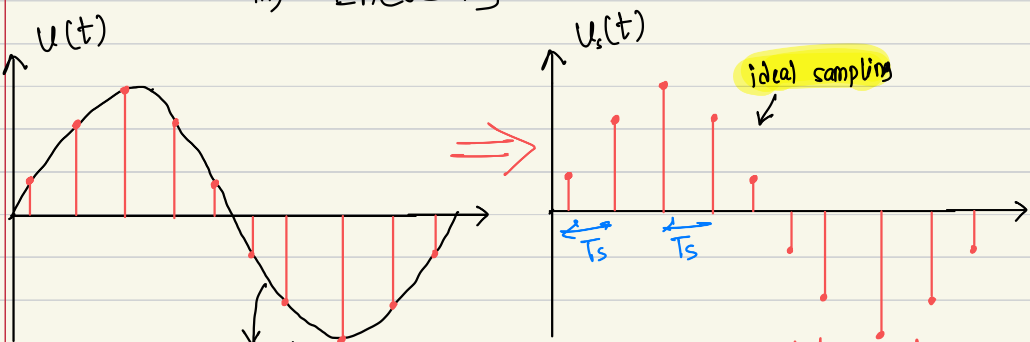
Digital data could take only two values (high or low). It means binary bit of strings. In order to store / process we use digital data in our computers and microprocessors.

Environment → Transducer → Computer → Speaker/Monitor

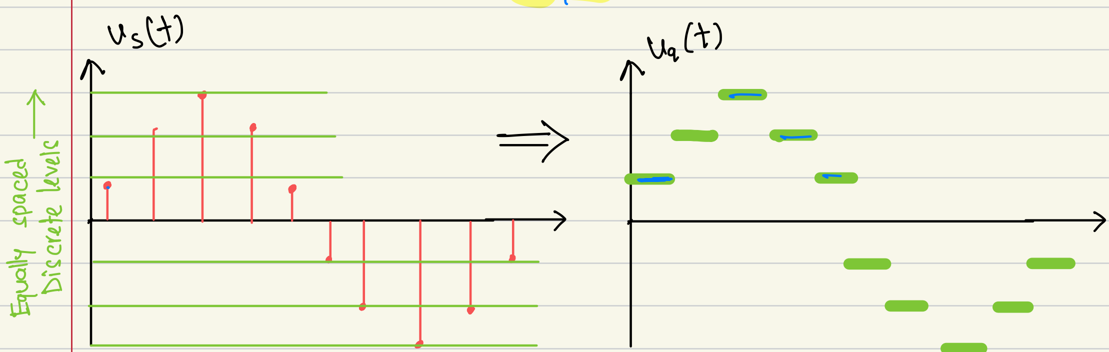
(Real world Analog Signal) (Analog i/p → Electrical o/p) ADC (Digital processing) DAC (Analog o/p)

A/D conversion process

→ It involves 3 basic steps
- i) Sampling
- ii) Quantization
- iii) Encoding



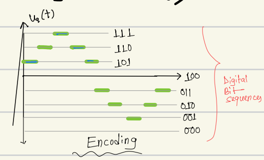
→ Sampling →



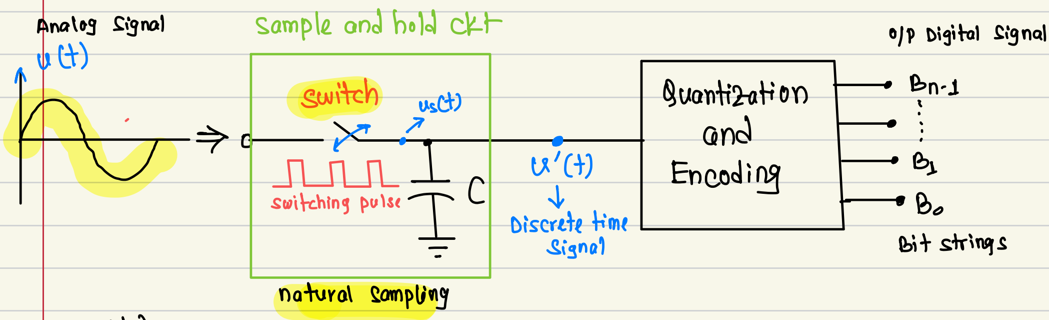
Sampled signal → Quantization → Quantized signal

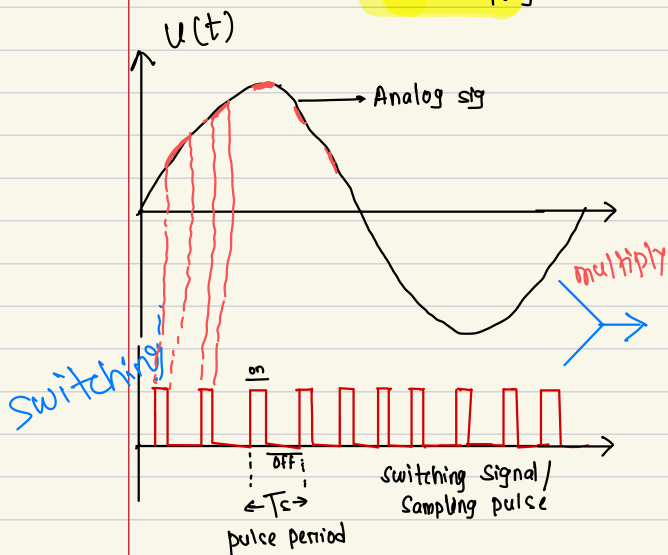
Quantized Signal $u_q(t)$ ⟹ Assigned to corresponding Binary values $\{u_e\}$ (Digital sig)

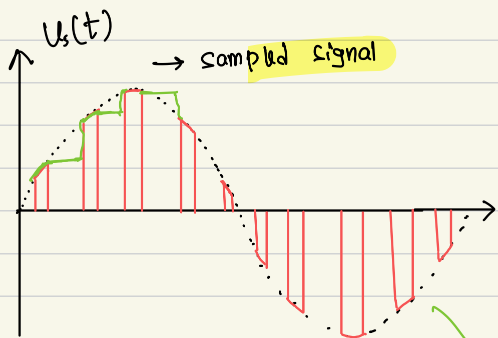
Encoding

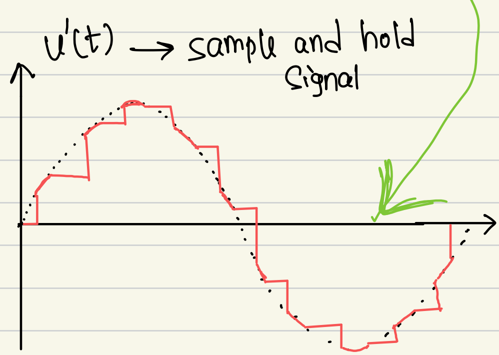


Practical implementation of ADC









This hold operation is done by using capacitor.

Quantization

↳ It is a process by which we assign sampled and hold signal data to some fixed preassigned values / levels.

→ for n bit quantization
num. of levels, $L = 2^n$
gaps btwn two levels → Resolution.

step size / → Resolution, $\Delta = \dfrac{V_{max} - V_{min}}{2^n}$

V_{max} → max value of Analog signal
V_{min} → min " " " "

$n\uparrow, L\uparrow, \Delta\downarrow$ → quantization error ↓

2 types

- i) Mid-rise type Quantization
- ii) Mid-tread type Quantization.

Formulas

given → n, V_{max}, V_{min}

→ levels, $L = 2^n$

→ stepsize / resolution, $\Delta = \dfrac{V_{max} - V_{min}}{2^n}$ → 1 LSB value.

→ Quantization error $= \dfrac{\Delta}{2}$

$$= \frac{1}{2} \frac{(V_{max} - V_{min})}{2^n}$$

→ Sampling period → T_s

→ Sampling Frequency, $F_s = \dfrac{1}{T_s}$

→ Minimum Sampling Frequency, $\boxed{F_s = 2 \times F_{input}}$

→ for n-bit Flash ADC

Hardwares

- → num of Resistors $= 2^n$
- → num of Comparators $= 2^n - 1$
- → $2^n : n$ priority Encoder

Made with Goodnotes

PP-2

An analog signal $8 \sin 4000\pi t$ is to be converted into Digital signal with a quantization error less than or equal to 0.781% of the input voltage range.

i) Find the required number of bits

ii) Determine the minimum sampling frequency and sampling Period.

Ans:

(i) We know $q.error = \frac{\Delta}{2} = \frac{1}{2} \frac{(V_{max} - V_{min})}{2^n}$

given $q.error = 0.781\% \ (V_{max} - V_{min})$

$$\therefore \frac{1}{2} \frac{(\cancel{V_{max} - V_{min}})}{2^n} = \frac{0.781}{100} (\cancel{V_{max} - V_{min}})$$

$$\frac{1}{2^{n+1}} = \frac{0.781}{100}$$

$$\therefore \boxed{n = 6}$$

(ii) $2\pi f_{i/p} t = 4000\pi t$

$\therefore f_{i/p} = 2000 \text{ Hz}$

$\therefore$ min sampling frequency $\rightarrow f_s = 2 f_{i/p}$

$$\boxed{f_s = 4000 \text{ Hz}}$$

Sampling Period, $T_s = \frac{1}{f_s} = \frac{1}{4000}$ sec

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